



Technical Service Bulletin

Technical Service Bulletin: TSB110025

Released Date: 08-Jun-2011

New Cylinder Head Capscrew and Gasket

New Cylinder Head Capscrew and Gasket

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Publications Affected	
Engine	Parts Catalogs
M11	3672021, 3672081, 3672082
L10	3379876, 3672109, 3822121, 3884215, 3884307
10G	3672049
ISM	3672143

This document introduces new vented, short intake port cylinder head capscrews (see Fig. 2), which went into production for the L10, L10G, M11, QSM, and ISM Series engines on March 27, 2000, with engine serial number first (ESN) 35000160. This document also introduces an improved steel carrier water grommet in the head gasket (see Fig. 3), which went into production for the L10, M11, and ISM engines on Mar6, 2000, with ESN first 34997974.

The new, short vented capscrews have an increased diameter flange with four radial grooves in the flange to vent the cylinder head bolt hole and prevent corrosion. The release of the steel carrier grommet enhances coolant-sealing characteristics. The new head gasket Part Number is 4022500.

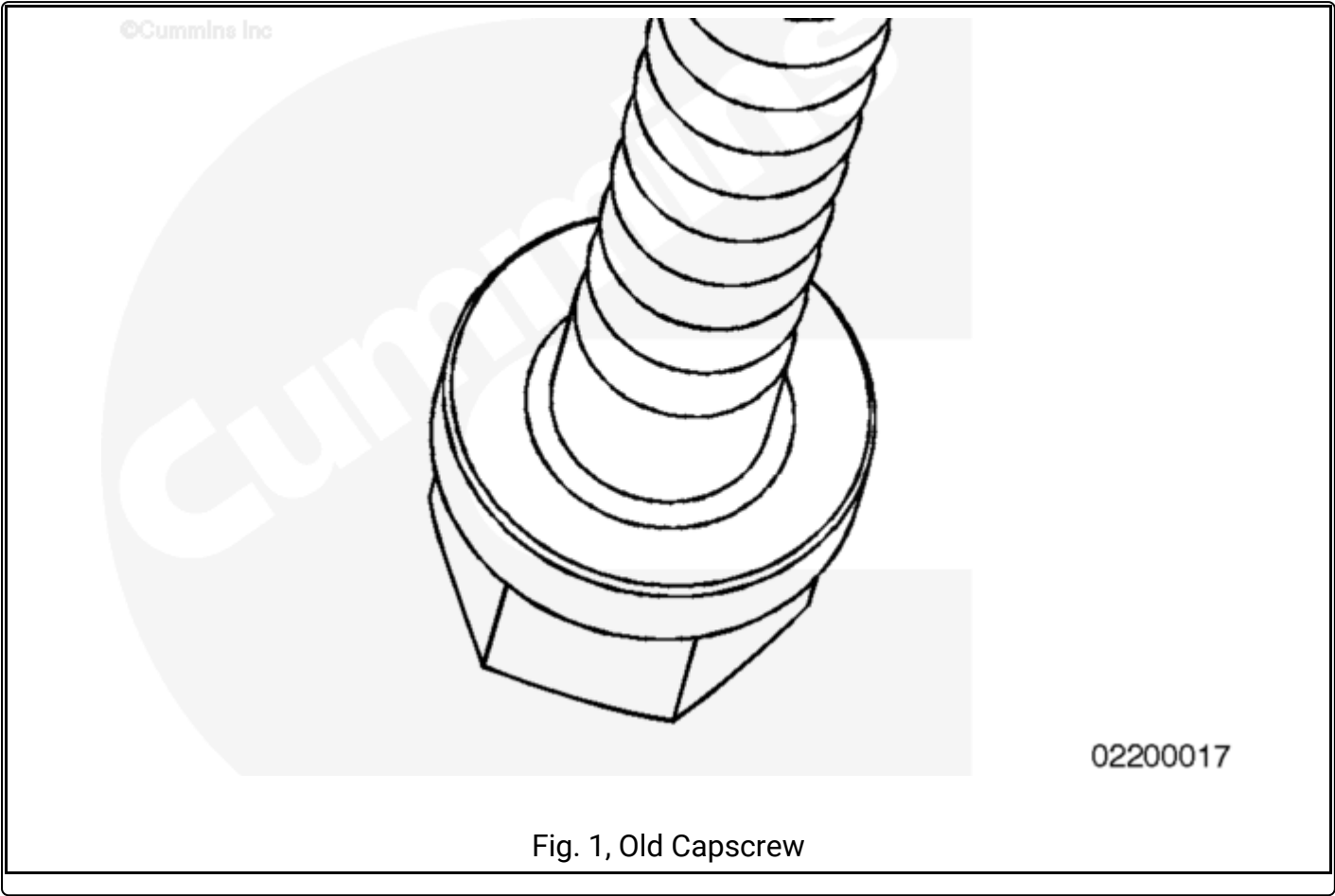
Table 1, Part Number Supersession

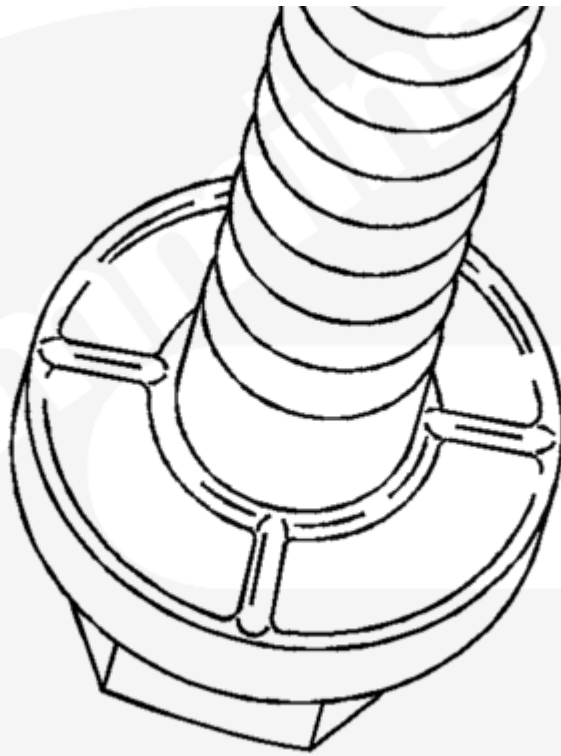
Old Part Number	New Part Number	Item Name
3161515	4022907	Capscrew
3328925	4022500	Cylinder head gasket

The upper engine gasket kit will include three new capscrews. The upper engine gasket kits will be superseded as listed in Table 2. Other kits will contain the improved head gasket with no part number changes.

Table 2, Upper Engine Gasket Set		
Old Part Number	New Part Number	Item Name
3800861	3800861	No
3803293	4024993	Yes
3804842	3804842	No
3804843	4024992	Yes
3804845	4024992	Yes
3806179	4024993	Yes

Note : This is a product improvement and is not subject to campaign.

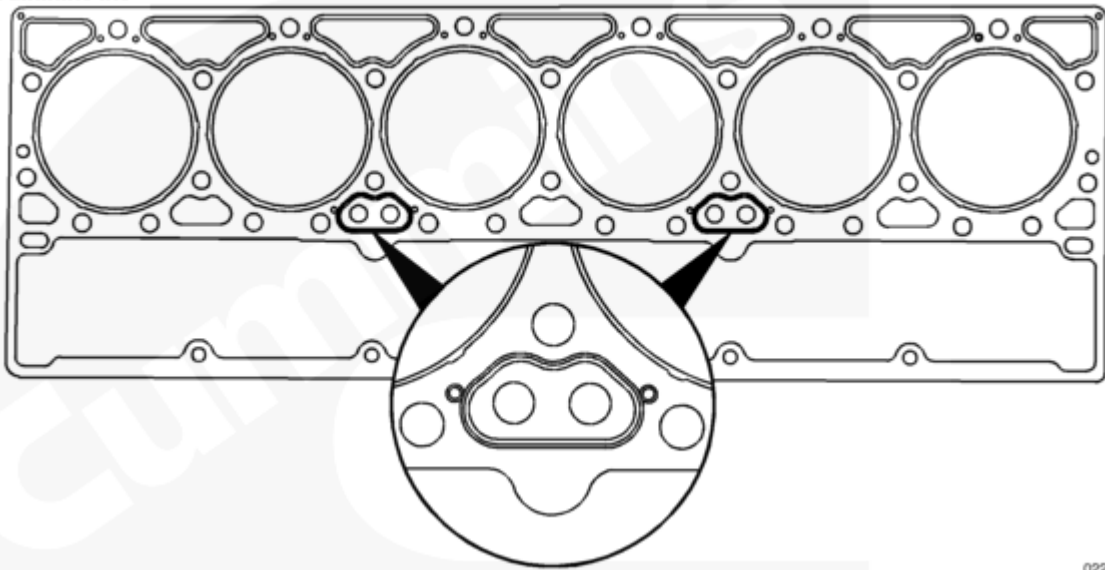




02200018

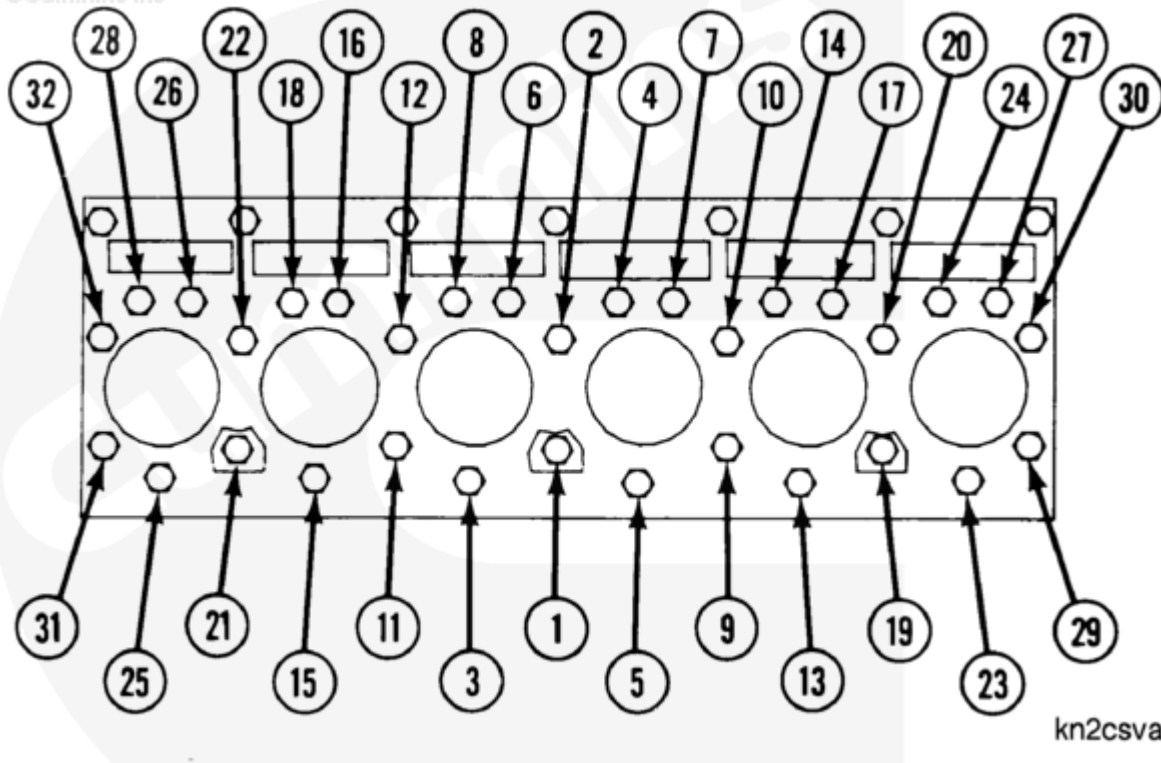
Fig. 2, New Vented Capscrew

The vented capscrews also can be ordered separately.



02200019

Fig. 3, Improved Steel Carrier Water Grommet



Note : Do not mix flange head capscrews and hex head capscrews with hardened washers on the same engine.

During head gasket repairs, use the present tightening sequence in the following procedure.

The old-style hex head capscrews and washers use the following installation procedure.

Tighten the capscrews in the sequence listed to the specified values.

Hex Head Capscrew Torque Value:

Step 1. 100 N•m [74 ft-lb]

Step 2. 170 N•m [125 ft-lb]

Step 3. 235 N•m [173 ft-lb]

Step 4. Loosen completely

Step 5. Repeat Steps 1 through 3.

Flange Head Capscrew Torque Value:

Step 1. 68 N•m [50 ft-lb]

Step 2. 136 N•m [100 ft-lb]

Step 3. Rotate 180 degrees.

Note : Rotate at least two flats, but not more than three flats.

Document History

Date	Details
2011-1-17	Module Created
2011-5-24	Format Error

Last Modified: 08-Jun-2011
